

Chapter 5

Error and Confidence Intervals

Every statistic we compute from a sample — a mean, a proportion — is an estimate. It is our best single guess at an unknown population parameter. But a single number without any sense of its reliability is incomplete. A doctor who says “the average recovery time is 12 days” is more useful when she adds “give or take about 2 days.”

This chapter formalizes that “give or take.” We will build **confidence intervals**: ranges of plausible values for a population parameter, constructed from sample data. By the end of this chapter, you will be able to say not just *what* you estimate, but *how confident* you are that the true value lies nearby.

The machinery comes entirely from Chapter 4: the CLT tells us the sampling distribution of $\bar{x}$, and the z -table lets us find the precise range that contains the true mean with any desired probability.

5.1 Point Estimates and Sampling Error

Definition: Point Estimate. A **point estimate** is a single number computed from sample data that serves as our best guess for a population parameter.

$$\bar{x} \text{ estimates } \mu \quad \hat{p} \text{ estimates } p$$

A point estimate is almost certainly wrong — not because of any mistake, but because samples vary. Different samples from the same population give different $\bar{x}$ values. The question is not whether the estimate is exactly right (it won’t be), but how far off it is likely to be.

Definition: Margin of Error. The **margin of error** (ME) is the maximum likely distance between the point estimate and the true parameter, at a specified confidence level. It reflects the precision of the estimate: smaller ME means more precision.

Definition: Confidence Interval. A **confidence interval** is an interval estimate of a population parameter:

$$\text{point estimate} \pm \text{margin of error.}$$

It gives a range of values that plausibly contains the true parameter.

5.2 The Logic of Confidence

Before writing any formulas, we need to understand what “95% confident” actually means. This is one of the most commonly misunderstood ideas in all of statistics.

5.2.1 What the Confidence Level Means

From Chapter 4: the CLT guarantees that $\bar{x} \sim N(\mu, \sigma/\sqrt{n})$. From Chapter 3 (and the Week 3 quiz!): $P(-1.96 \leq Z \leq 1.96) = 0.95$.

Putting these together: for 95% of all possible samples,

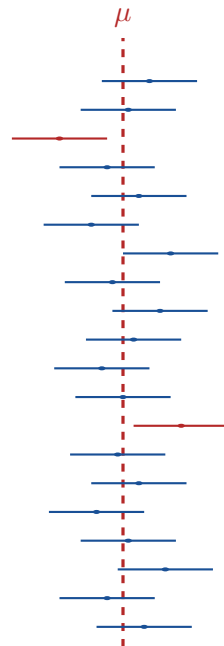
$$\mu - 1.96 \cdot \frac{\sigma}{\sqrt{n}} \leq \bar{x} \leq \mu + 1.96 \cdot \frac{\sigma}{\sqrt{n}}.$$

Rearranging (subtracting $\bar{x}$ and flipping inequalities):

$$\bar{x} - 1.96 \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}.$$

This is the 95% confidence interval for μ . It says: in 95% of all samples, the interval built this way will contain the true population mean μ .

Visual Insight. The picture below shows 20 different samples from the same population, each producing a different $\bar{x}$ and a different confidence interval. The vertical dashed line is the true (fixed, unknown) μ . Notice that most intervals contain μ — but not all. About 1 in 20 miss. That is what “95% confidence” means.



μ (unknown, fixed)

Blue intervals contain μ ; red intervals miss. 18 out of 20 hit (90%), close to 95%. With many more samples, exactly 95% would contain μ .

The key insight: μ does not move. The intervals move — each one centered at a different $\bar{x}$. “95% confidence” means the procedure, repeated many times, captures μ in 95% of the resulting intervals.

Common Pitfall. The most common misconception in statistics: “There is a 95% probability that μ is in this interval.”

This is wrong. Once the interval is computed, μ either is in it or it is not. There is no probability left — it is a fixed (unknown) number.

The correct statement: “I used a procedure that, in the long run, produces intervals containing μ 95% of the time. This particular interval was produced by that procedure.”

Exercises 5.2

5.2.1. A polling firm reports: “The candidate’s approval rating is 48%, with a margin of error of 3% at 95% confidence.” Write the confidence interval. What does the 95% mean in plain language?

5.2.2. In the picture above, if we had used 90% confidence instead of 95%, would

the intervals be wider or narrower? Would more or fewer of them miss μ ?

5.2.3. True or False: If I build a 95% CI and it is (42, 58), the probability that $\mu = 50$ is 95%. Explain your answer carefully.

5.3 Confidence Interval for a Population Mean (σ Known)

Key Formula. A z -confidence interval for μ (when σ is known and $n \geq 30$):

$$\bar{x} \pm z^* \cdot \frac{\sigma}{\sqrt{n}}$$

where z^* is the **critical value** from the standard normal table determined by the confidence level:

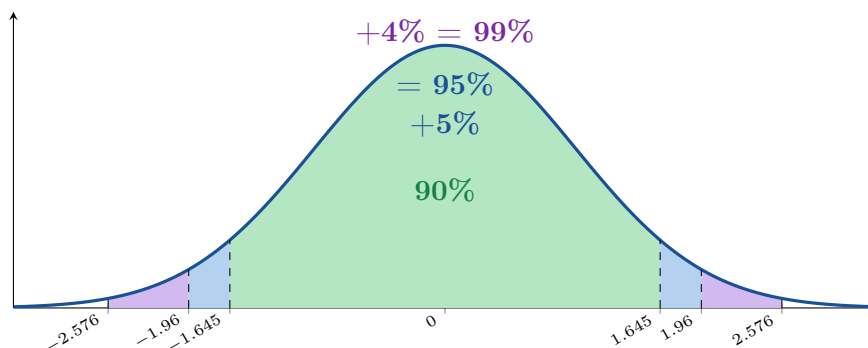
Confidence level	z^*	Area in each tail
90%	1.645	5%
95%	1.960	2.5%
99%	2.576	0.5%

The **margin of error** is $ME = z^* \cdot \frac{\sigma}{\sqrt{n}}$.

5.3.1 Where the Critical Values Come From

The value $z^* = 1.96$ captures 95% of the standard normal distribution between -1.96 and $+1.96$ — this is exactly what you found in the Week 3 quiz. For 90% confidence, we need the middle 90%, leaving 5% in each tail: $P(Z \leq z^*) = 0.95 \Rightarrow z^* = 1.645$.

Visual Insight. The three critical values on the same standard normal curve:

Critical values z^* for 90%, 95%, and 99% confidence

Higher confidence requires a wider net (z^* increases), which widens the interval. There is always a tradeoff: more confidence means less precision.

Example 5.3.1. (*95% CI for a Mean*) A nutritionist samples 40 adults and records their daily sodium intake. She finds $\bar{x} = 2,850$ mg. From prior research, $\sigma = 400$ mg. Construct a 95% confidence interval for the true mean daily sodium intake μ .

Step 1. Check conditions. $n = 40 \geq 30$. CLT applies. σ is known.

Step 2. Identify z^ .* 95% CI: $z^* = 1.96$.

Step 3. Compute ME.

$$\text{ME} = 1.96 \cdot \frac{400}{\sqrt{40}} = 1.96 \cdot 63.25 \approx 123.97 \approx 124 \text{ mg.}$$

Step 4. Build the interval.

$$2,850 \pm 124 \Rightarrow (2,726, 2,974) \text{ mg.}$$

Interpretation. We are 95% confident that the true mean daily sodium intake for adults lies between 2,726 and 2,974 mg.

Example 5.3.2. (*Effect of Confidence Level*) Using the same data ($\bar{x} = 2,850$, $\sigma = 400$, $n = 40$), construct a 99% confidence interval.

$z^* = 2.576$, $\text{ME} = 2.576 \cdot 63.25 \approx 163$ mg.

99% CI: (2,687, 3,013) mg.

Compare to the 95% interval (2,726, 2,974). The 99% interval is wider — more confidence costs precision.

5.3.2 Effect of Sample Size on Margin of Error

Since $\text{ME} = z^* \cdot \sigma / \sqrt{n}$, increasing n decreases ME. But the relationship is not linear: to *halve* the margin of error, you must *quadruple* the sample size (because $1/\sqrt{4n} = \frac{1}{2} \cdot 1/\sqrt{n}$).

Key Formula. Required sample size to achieve margin of error $\leq E$:

$$n \geq \left(\frac{z^* \cdot \sigma}{E} \right)^2.$$

Always round *up* to the nearest whole number.

Example 5.3.3. (*Required Sample Size*) A researcher wants a 95% CI for mean height with margin of error no more than 0.5 inches. Historical data suggest $\sigma = 3$ inches. How large a sample is needed?

$$n \geq \left(\frac{1.96 \cdot 3}{0.5} \right)^2 = \left(\frac{5.88}{0.5} \right)^2 = (11.76)^2 = 138.3.$$

Round up: $n \geq 139$ people.

Exercises 5.3

- 5.3.1. A sample of 50 light bulbs has $\bar{x} = 1,182$ hours. Assume $\sigma = 80$ hours.
 (a) Construct a 90% CI for μ . (b) Construct a 99% CI for μ . (c) Which interval is wider? Which gives more precision?
- 5.3.2. Same problem. How many bulbs would need to be sampled to achieve a 95% CI with margin of error no more than 15 hours?
- 5.3.3. Explain in one sentence why $z^* = 1.96$ for 95% confidence but $z^* = 1.645$ for 90% confidence. (Draw the curve if it helps.)

5.4 Confidence Interval for a Population Proportion

Many questions are about proportions: what fraction of voters support a candidate, what fraction of patients respond to a drug, what fraction of parts are defective.

5.4.1 The Sampling Distribution of $\hat{p}$

By the same CLT logic applied to counts instead of measurements:

Key Formula. For a sample proportion $\hat{p} = x/n$ (where x is the count of suc-

cesses):

$$\hat{p} \approx N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

provided $np \geq 10$ and $n(1-p) \geq 10$ (the **success-failure condition**).

The standard deviation of $\hat{p}$ is $\sqrt{p(1-p)/n}$. Since p is unknown, we replace it with $\hat{p}$ to get the **standard error of $\hat{p}$** :

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Key Formula. A confidence interval for a population proportion p :

$$\hat{p} \pm z^* \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Conditions: $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$.

Example 5.4.1. (CI for a Proportion) In a survey of 320 college students, 176 reported eating breakfast every day. Construct a 95% confidence interval for the true proportion of college students who eat breakfast daily.

Step 1. Compute $\hat{p}$. $\hat{p} = 176/320 = 0.55$.

Step 2. Check conditions. $n\hat{p} = 320(0.55) = 176 \geq 10$. $\checkmark n(1-\hat{p}) = 320(0.45) = 144 \geq 10$. $\checkmark$

Step 3. Compute SE and ME.

$$SE = \sqrt{\frac{0.55 \cdot 0.45}{320}} = \sqrt{\frac{0.2475}{320}} = \sqrt{0.000773} \approx 0.0278.$$

$$ME = 1.96 \cdot 0.0278 \approx 0.0545.$$

Step 4. Build the interval.

$$0.55 \pm 0.0545 \Rightarrow (0.496, 0.605).$$

Interpretation. We are 95% confident that between 49.6% and 60.5% of college students eat breakfast every day.

Common Pitfall. For the proportion CI, the standard error uses $\hat{p}$, not p (which is unknown). The formula $\sqrt{\hat{p}(1-\hat{p})/n}$ looks similar to the binomial formula $\sqrt{p(1-p)/n}$, but we are estimating the unknown p with $\hat{p}$. Always use $\hat{p}$ in the standard error when constructing the CI.

Exercises 5.4

5.4.1. A random sample of 500 voters found that 285 plan to vote for Candidate A. (a) Compute $\hat{p}$. (b) Check the success-failure condition. (c) Construct a 95% CI for the true proportion p . (d) Based on this interval, can you conclude the majority (more than 50%) will vote for Candidate A? Explain.

5.4.2. A 90% CI for a proportion is $(0.34, 0.46)$. What is $\hat{p}$? What is the margin of error?

5.4.3. How does the standard error of $\hat{p}$ behave when $\hat{p} = 0.5$ compared to $\hat{p} = 0.1$? When is estimation most uncertain?

5.5 Interpreting and Communicating Confidence Intervals

A confidence interval is only useful if its interpretation is correct. Here we assemble all the pieces.

5.5.1 The Four-Step Process

Every confidence interval problem should follow four steps:

1. **State:** Identify the parameter. State the confidence level. Check conditions (CLT applies? Success-failure?).
2. **Estimate:** Compute the point estimate ($\bar{x}$ or $\hat{p}$).
3. **Margin of Error:** Compute z^* and ME.
4. **Interpret:** Write the interval and interpret it in context. Use the phrase: “We are __% confident that the true [parameter] is between [lower] and [upper].”

5.5.2 What Affects the Width of a Confidence Interval

If you increase...	Width changes	Why
Confidence level (z^*)	Wider	Larger z^*
Sample size n	Narrower	Larger $\sqrt{n}$ in denominator
Population spread σ	Wider	More variability to capture

Visual Insight. The tradeoff between confidence and width is fundamental. Imagine catching a moving target with a net. A wide net is more likely to catch the target — but it tells you less precisely where the target is. A narrow net may miss the target, but if it catches it, you know exactly where the target is.

A 99% confidence interval is a wide net. A 90% interval is a narrow net. You choose the width based on what you are willing to risk and what precision you need.

Chapter 5 Review

1. A sample of 64 student loan balances yields $\bar{x} = \$28,400$ with known $\sigma = \$6,200$.
 (a) Construct a 90% CI for the true mean balance μ . (b) Construct a 95% CI.
 (c) Construct a 99% CI. (d) Compare the three intervals: what is the tradeoff?
2. A university wants to estimate the mean study time per week for its students to within 1 hour using a 95% CI. Prior research suggests $\sigma = 5$ hours. How many students must be sampled?
3. A random sample of 400 adults found that 148 reported getting 8 or more hours of sleep per night. (a) Compute $\hat{p}$. Check conditions. (b) Construct a 95% CI for the true proportion p . (c) Interpret the interval in context. (d) A health agency claims 40% of adults get enough sleep. Is this claim consistent with your interval? Explain.
4. A 95% CI for a mean is computed as (44.2, 55.8). (a) What is $\bar{x}$? What is the margin of error? (b) Would a 90% CI from the same data be wider or narrower? Write the approximate interval without recomputing. (c) A friend says: “There is a 95% chance the true mean is 50.” Correct this statement.
5. *Big picture:* In three to four sentences, explain how the Central Limit Theorem (Chapter 4) makes confidence intervals (this chapter) possible. What would break down if the CLT did not hold?